

# Mubashir Rehman

ERP + AI Engineer · Frappe / ERPNext · Django HRMS · LLM Agents & Business Automation · Python

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## SUMMARY

Lead Software Engineer with **3+ years** building on systems of record: Frappe/ERPNext and Django HRMS platforms, AI agent layers on top of them, and the Python and PostgreSQL backends and deployment around them. Co-author of a peer-reviewed paper in Springer's Circuits, Systems, and Signal Processing (2025).

## PUBLICATION

**Evaluation of a low-cost single-lead ECG module for vascular ageing prediction and studying smoking-induced changes in ECG**

*Circuits, Systems, and Signal Processing (Springer)*, 44(8), 5852-5875, 2025 · DOI: 10.1007/s00034-025-03048-2

Co-author, equal contribution (3rd of 7 authors): data collection and the ML pipeline. **42 subjects, 6,131 segmented samples**; random forest **R<sup>2</sup> 0.99** on segmented data and **0.87** with transfer learning on unsegmented data.

## EXPERIENCE

**Lead Software Engineer, TransData** · Lahore, Pakistan · Mar 2025 to Present (Lead since Mar 2026)

- Built an AI-agent ERP platform that grew from expense-claim automation into multi-company procurement (**two live deployments, demo and production**), by writing LLM agents with forced tool-calling against ERPNext through one versioned contract, a multi-company resolver, and purchase-order drafting and approval with audit logging.
- Consolidated HR, CRM and projects onto one whitelabeled Frappe/ERPNext platform, **sole author (18 of 18 commits), live deployment**, replacing a legacy HRMS, Excel CRM tracking and Asana, by building the HR and Asana project migrations, custom doctypes and scripts, and the Docker deployment with biometric attendance sync.
- Delivered, as sole engineer, a customised Django HRMS covering **138 employee records and ZKTeco biometric sync across 2 devices** (since replaced by the Frappe platform above), by designing a multi-device attendance schema, reporting endpoints, payroll calculation logic, and Celery jobs.
- Implemented **10+ of 28 spec'd AI features** (27 of 27 commits, sole author) for a property-management platform, by building tenant scoring, churn and renewal prediction, sentiment analysis, document classification and LLM reply drafting, live-tested against the OpenAI API.
- Built and hardened an AI voice agent layered onto a dental practice-management system through its **production launch**, by wiring voice agents to booking and eligibility tools and fixing the practice-management sync.
- Stood up a Frappe CRM demo for EU prospects, **kept live as a demo deployment** (3 of 4 commits), by configuring an EU company, hand-building the demo data Frappe no longer ships, and hardening the asset image build.

**Software Engineering Team Lead, VeritusLabs** · Oct 2023 to Mar 2025 (Software Engineer Jul 2023 to Oct 2023)

- Promoted to Team Lead and led **a team of 4 engineers** on QGroundControl-based drone-control and swarm features in C++ and Qt/QML, including mission planning and real-time telemetry over MQTT.
- Delivered a full-stack AI prompt-management web app with streaming OpenAI responses, RBAC and S3-backed storage, **deployed through Docker and CI/CD**, by building its Django/FastAPI and LangChain backend.

## PROJECTS

**HireTrack** · React 19, TypeScript, LangGraph, Supabase · github.com/mubashir-rehman/job-application-tracker

- Built and open-sourced a job-application tracker (sole author, **89 commits, public repo, live deployment**), by shipping a deterministic-first LangGraph pipeline that tailors resumes and scores job descriptions across five AI providers (bring-your-own-key), plus an offline-first PWA and a remote MCP server.

## SKILLS

**ERP and business systems:** ERPNext/Frappe (custom doctypes, server and client scripts, bench CLI, REST API), Frappe CRM, Horilla HRMS, ZKTeco biometric API, HL7 integration

**AI/LLM:** OpenAI API, Gemini, LangChain, LangGraph, agent tool-calling, MCP, embeddings, pgvector

**Backend and data:** Python, SQL, JavaScript, TypeScript, Django, Django REST Framework, FastAPI, Celery, PostgreSQL, MySQL, Redis, REST API design, OAuth2, JWT, RBAC

**DevOps:** Docker, Nginx, Traefik, GitHub Actions, CI/CD, AWS, DigitalOcean, Linux, Systemd

## EDUCATION

**BS Computer Science**, Information Technology University, Lahore